Practical 3

In [3]:
```
import pandas as pd
import numpy as np
```

In [4]:
```
df=pd.read_csv("titanic.csv")
```

In [8]:
```
df.isnull().sum()
```

Out[8]:
```
PassengerId      0
Survived         0
Pclass           0
Lname            0
Name             0
Sex              0
Age             30
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          125
Embarked         1
dtype: int64
```

In [9]:
```
df["Age"].mean()
```

Out[9]:
```
np.float64(28.141507936507935)
```

In [10]:
```
df["Age"]=df["Age"].fillna(df["Age"].mean()).round(2)
```

In [13]:
```
df.isnull().sum()
```

Out[13]:
```
PassengerId      0
Survived         0
Pclass           0
Lname            0
Name             0
Sex              0
Age              0
SibSp            0
Parch            0
Ticket           0
Fare             0
Cabin          125
Embarked         1
dtype: int64
```

In [14]:
```
grouped_data=df.groupby('Sex')['Age']
```

In [16]:
```
summary=grouped_data.describe()
```

In [17]:
```
summary
```

Out[17]:

	count	mean	std	min	25%	50%	75%	max
Sex								
female	56.0	25.058214	11.903435	2.00	17.0	27.00	29.25	58.0
male	100.0	29.867700	13.511047	0.83	22.0	28.14	34.25	71.0

In [18]:
```
summary_2=grouped_data.agg('mean','median')
```

In [19]:
```
summary_2
```

Out[19]:
```
Sex
female    25.058214
male      29.867700
Name: Age, dtype: float64
```

In [20]:
```
age_list=grouped_data.apply(list)
```

In [21]:
```
age_list
```

Out[21]:
```
Sex
female    [38.0, 26.0, 35.0, 27.0, 14.0, 4.0, 58.0, 14.0...
male      [22.0, 35.0, 28.14, 54.0, 2.0, 20.0, 39.0, 2.0...
Name: Age, dtype: object
```

In [22]:
```
df2=pd.read_csv("iris.csv")
```

In [23]:
```
df2.isnull().sum()
```

Out[23]:
```
sepal_length    0
sepal_width     0
petal_length    0
petal_width     0
species         0
dtype: int64
```

In [32]:
```
grouped_iris=df2.groupby('species')
```

In [33]:
```
summary_iris=grouped_iris.describe()
```

In [34]:
```
summary_iris
```

Out[34]:

	sepal_length								sepal_width		...	pet
	count	mean	std	min	25%	50%	75%	max	count	mean	...	75
species												
Iris-setosa	50.0	5.006	0.352490	4.3	4.800	5.0	5.2	5.8	50.0	3.418	...	1.5
Iris-versicolor	50.0	5.936	0.516171	4.9	5.600	5.9	6.3	7.0	50.0	2.770	...	4.6
Iris-virginica	50.0	6.588	0.635880	4.9	6.225	6.5	6.9	7.9	50.0	2.974	...	5.8

3 rows × 32 columns

In [36]:

```
for spec,data in grouped_iris:
    print("Species=",spec)
    print(data.describe())
```

```
Species= Iris-setosa
       sepal_length  sepal_width  petal_length  petal_width
count      50.00000    50.000000     50.000000     50.00000
mean        5.00600     3.418000      1.464000      0.24400
std         0.35249     0.381024      0.173511      0.10721
min         4.30000     2.300000      1.000000      0.10000
25%         4.80000     3.125000      1.400000      0.20000
50%         5.00000     3.400000      1.500000      0.20000
75%         5.20000     3.675000      1.575000      0.30000
max         5.80000     4.400000      1.900000      0.60000
Species= Iris-versicolor
       sepal_length  sepal_width  petal_length  petal_width
count     50.000000    50.000000     50.000000    50.000000
mean       5.936000     2.770000      4.260000     1.326000
std        0.516171     0.313798      0.469911     0.197753
min        4.900000     2.000000      3.000000     1.000000
25%        5.600000     2.525000      4.000000     1.200000
50%        5.900000     2.800000      4.350000     1.300000
75%        6.300000     3.000000      4.600000     1.500000
max        7.000000     3.400000      5.100000     1.800000
Species= Iris-virginica
       sepal_length  sepal_width  petal_length  petal_width
count      50.00000    50.000000     50.000000     50.00000
mean        6.58800     2.974000      5.552000      2.02600
std         0.63588     0.322497      0.551895      0.27465
min         4.90000     2.200000      4.500000      1.40000
25%         6.22500     2.800000      5.100000      1.80000
50%         6.50000     3.000000      5.550000      2.00000
75%         6.90000     3.175000      5.875000      2.30000
max         7.90000     3.800000      6.900000      2.50000
```

In [ ]: